AEGIS: Mining Graph Structure for Adversarial Vulnerability Analysis of GNNs

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Abstract—**Adversarial manipulation of graph structure poses a growing threat as graph neural networks are deployed in fraud detection, drug interaction prediction, and infrastructure monitoring. Yet practitioners lack tools to answer a basic question before deployment: *which edges, if perturbed, would flip this model's predictions?* We introduce AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity), a framework that answers this question by constructing a constrained sensitivity matrix S_c mapping edge perturbations to hidden-state shifts. From S_c, AEGIS extracts the mathematically optimal attack direction (via SVD), per-edge vulnerability rankings, and per-node first-order sensitivity radii. A matrix-free formulation based on Neumann-series resolvent iteration and randomized SVD enables full-graph analysis (up to N=7,650 on a single GPU; Cora in 78 s / 1 GB, Amazon Photo in 363 s / 6.5 GB) without materializing any $Nd \times N^2$ matrix. The computation applies to any GNN whose message passing is differentiable with respect to continuous edge weights, serving as a practical vulnerability analysis tool; for contractive implicit models (IGNN-class), it additionally provides formal guarantees including a critical perturbation budget $\varepsilon_{\mathrm{crit}}$ and a three-regime vulnerability characterization. Under realistic (symmetric, edge-only) perturbations, first-order predictions achieve tightness 1.00 ± 0.01 at $\varepsilon = 0.01$ and remain within 15% at $\varepsilon = 0.10$ across 7 architectures and 9 datasets. The SVD-optimal attack inflicts 2–8× more damage than random perturbation. On IEEE power grids, the vulnerability spectrum recovers N-1 contingency rankings (P@10 = 0.66–0.81) without requiring line-impedance data, demonstrating that structural sensitivity analysis can serve as a pre-deployment screening tool. Code will be released upon publication.**

Index Terms—**Graph neural networks, adversarial robustness, structural sensitivity, deep equilibrium models, graph perturbation, implicit function theorem**

I. INTRODUCTION

Graph neural networks have entered domains where incorrect predictions carry tangible consequences: fraudulent accounts evade detection when an adversary inserts edges into a financial transaction graph [1], drug interaction models produce unsafe recommendations when molecular graphs are perturbed [2], and power grid monitoring systems may miss critical contingencies if their graph-based representations are structurally fragile [3]. In all these settings, the graph topology is not merely an input format but a potential attack surface. A practitioner preparing to deploy a GNN faces a fundamental question that current tools leave unanswered: *which edges in this graph, if perturbed, would cause the model's predictions to fail, and by how much?*

This question is distinct from both adversarial attack and certified defense. Attack methods such as Nettack [1] and Mettack [4] search for damaging perturbations using surrogate gradients or meta-learning, but they provide no guarantee that the perturbation found is optimal, and they do not quantify per-node vulnerability. Certified defenses based on randomized smoothing [5], [6] or interval bound propagation [7], [8], [9] provide robustness guarantees, but they are uniform: every node receives the same certificate, revealing nothing about *which* edges or nodes are structurally weak. Empirical defenses [10], [11], [12], [13] harden the model but offer no formal understanding of residual vulnerability. What is missing is a *structural vulnerability map* that, given a trained GNN and a graph, identifies the optimal perturbation direction, the weakest edges, and the perturbation tolerance of each node.

We introduce AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity), a framework that produces exactly this map. The central object is the *constrained sensitivity matrix* $S_c \in \mathbb{R}^{Nd\times|E|}$, which projects the full N^2-dimensional adjacency perturbation space onto the $|E|$-dimensional space of realistic (symmetric, edge-only) perturbations. From S_c, three outputs follow directly: (i) the SVD yields the first-order optimal attack direction, (ii) column norms give per-edge vulnerability scores, and (iii) the classification margin divided by the sensitivity norm gives per-node tolerance radii. The S_c computation applies to any GNN whose message passing is differentiable with respect to continuous edge weights (theorem 5), yielding vulnerability rankings, SVD-optimal attacks, and first-order sensitivity radii as a practical computational tool. For contractive implicit GNNs (IGNN-class [14]), the implicit function theorem additionally provides formal guarantees: a critical perturbation budget $\varepsilon_{\mathrm{crit}} = (1-\kappa)/\left\|W\right\|_2$ and a three-regime characterization of model behavior under increasing perturbation (theorem 1). These formal guarantees are specific to the implicit case and do not transfer to explicit architectures.

These predictions are quantitatively accurate: constrained first-order tightness is 1.00 ± 0.01 at $\varepsilon = 0.01$ and remains within 15% at $\varepsilon = 0.10$, across 7 GNN architectures and 9 datasets spanning citation networks, e-commerce, encyclopedias, and power grids. On IEEE power grids, the vulnerability spectrum independently recovers N-1 contingency rankings (P@10 = 0.66–0.81), demonstrating that structural sensitivity analysis transfers to real-world engineering problems without domain-specific modifications.

Contributions.

1) **Constrained sensitivity framework.** We derive the constrained sensitivity matrix S_c for any GNN whose message passing is differentiable with respect to edge weights (theorem 5). For contractive implicit models (IGNN-class), we additionally prove a critical budget

$\varepsilon_{\text{crit}}$ and three vulnerability regimes (theorem 1); these formal guarantees do not transfer to explicit architectures.

2) **Scalable matrix-free computation.** A Neumann-series resolvent combined with autograd JVPs and randomized SVD [15] computes S_c's action without materializing any $Nd \times N^2$ matrix, enabling full-graph analysis (Cora N=2,708 in 78 s). We validate SVD-optimal attacks (2–8× stronger than random), per-edge vulnerability rankings, and per-node sensitivity radii across 7 architectures (section V-H).
3) **Cross-domain validation and case study.** We validate across 9 datasets in 4 domains, including a power grid case study where AEGIS recovers N-1 contingency rankings (τ between continuous S_c scores and discrete edge-removal ground truth: $+0.32$–$+0.54$; section VI).

II. Preliminaries and Threat Model

A. Notation and Formalization

Let $G = (V, E, A)$ denote an undirected graph with $N = |V|$ nodes, edge set E, and adjacency matrix $A \in \mathbb{R}^{N\times N}$. Each node $v \in V$ carries a feature vector $x_v \in \mathbb{R}^{d_{\text{in}}}$, collected into $X \in \mathbb{R}^{N\times d_{\text{in}}}$. We write $\hat{A} = D^{-1/2}(A + I)D^{-1/2}$ for the symmetric normalized adjacency with self-loops, where D is the degree matrix of $A + I$. Throughout, ϕ denotes the activation function, $\sigma_i(\cdot)$ denotes the i-th singular value, and $\text{vec}(\cdot)$ stacks a matrix column-wise into a vector (with reshape as its inverse). table I summarizes the key sensitivity objects.

Deep equilibrium GNNs. A DEQ-GNN [16], [17] defines a weight-tied operator $F_\theta : \mathbb{R}^{N\times d} \times \mathbb{R}^{N\times N} \to \mathbb{R}^{N\times d}$ and iterates it from $Z_0 = 0$ until convergence:

$$Z_{k+1} = F_\theta(Z_k, A), \quad z^* = \lim_{k\to\infty} Z_k. \tag{1}$$

When F_θ is contractive in operator norm ($\kappa = \|J_z\|_2 < 1$, where $J_z = \partial F/\partial z|_{z^*}$), the Banach fixed-point theorem guarantees a unique equilibrium $z^* = F_\theta(z^*, A)$ regardless of initialization.

IGNN instantiation. The Implicit Graph Neural Network (IGNN) [14] uses the operator

$$F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}}), \tag{2}$$

where $W \in \mathbb{R}^{d\times d}$ is the shared weight matrix, $X_{\text{proj}} \in \mathbb{R}^{N\times d}$ is a learned feature projection, and ϕ is ReLU activation. Contractivity is enforced by constraining $\|W\|_2$ via spectral normalization [18], ensuring $\kappa \le \left\|\hat{A}\right\|_2 \cdot \|W\|_2 < 1$. Training uses implicit differentiation through the fixed point [19], [20], and Jacobian regularization [21] can further stabilize convergence. A linear classification head $f(z^*) = W_{\text{cls}}z^* + b$ maps the equilibrium to class predictions.

Implicit function theorem. At the equilibrium, the equation $G(z^*, A) = z^* - F(z^*, A) = 0$ defines z^* as an implicit function of A. The IFT gives the sensitivity of z^* to any parameter p:

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1}\frac{\partial F}{\partial p}\bigg|_{z^*}, \tag{3}$$

TABLE I: Key sensitivity objects in AEGIS.

Symbol	Size	Description
S	$Nd \times N^2$	Unconstrained sensitivity matrix
S_c	$Nd \times \|E\|$	Constrained (symmetric, edge-only)
S_K	$Nd \times N^2$	Unrolled sensitivity for K-layer GNN
S_v	$d \times N^2$	Block-rows of S for node v
κ	scalar	Operator-norm contractivity $\|J_z\|_2$
ρ	scalar	Spectral radius $\rho(J_z)$
η	scalar	Pseudospectral index ≥ 1
$\varepsilon_{\text{crit}}$	scalar	Critical perturbation budget
r_v	scalar	Per-node first-order sensitivity radius

where the resolvent $(I - J_z)^{-1}$ exists and satisfies $\left\|(I - J_z)^{-1}\right\|_2 \le 1/(1 - \kappa)$ by the Neumann series (convergent because $\kappa < 1$). This bound uses the operator norm $\kappa = \|J_z\|_2$; the weaker spectral-radius bound $1/(1 - \rho)$ holds only for normal J_z. The pseudospectral index $\eta = \left\|(I - J_z)^{-1}\right\|_2 \cdot (1 - \rho) \ge 1$ [22] quantifies this gap, with $\eta = 1$ for normal matrices. The IFT has been used for hyper-parameter optimization [23] and implicit differentiation [19], but not for adversarial structural analysis.

B. Threat Model

We consider a white-box attacker who has full access to the trained IGNN model and perturbs the normalized adjacency matrix:

$$\hat{A}' = \hat{A} + \delta\hat{A}, \quad \text{subject to} \quad \left\|\delta\hat{A}\right\|_F \le \varepsilon. \tag{4}$$

The perturbation is further constrained to be:

- **Symmetric:** $\delta\hat{A} = \delta\hat{A}^\top$ (preserving the undirected nature of the graph).
- **Edge-only:** $[\delta\hat{A}]_{ij} = 0$ for $(i, j) \notin E$ (only existing edge weights may be modified; no new edges are introduced).
- **Continuous:** edge weights are perturbed continuously in $\mathbb{R}$, not discretely flipped.

This threat model captures continuous edge-weight perturbations to the normalized message-passing operator. It aligns naturally with the IGNN architecture, where $\hat{A}$ appears directly in F (eq. (2)). Discrete edge insertions or deletions, which change the graph topology and require recomputing the degree normalization $D^{-1/2}$, are outside the formal guarantee.

The key distinction from input-feature perturbation (studied for DEQs in [24], [25]) is that structural perturbation changes the *operator itself*: the equilibrium depends on A through both the state Jacobian $J_z = \text{diag}(\phi') \cdot (\hat{A} \otimes W)$ and the adjacency Jacobian $J_A = \partial F/\partial \text{vec}(A)$. This dual dependence makes structural sensitivity qualitatively different from input sensitivity.

C. Problem Statement

Given a trained contractive implicit GNN with equilibrium z^* on graph G, we aim to answer three questions:

1) **Vulnerability ranking:** Which edges, if perturbed, cause the largest shift in the equilibrium? That is, find a

ranking of edges $(i,j) \in E$ by their adversarial impact on z^*.
2) **Optimal attack:** What is the worst-case perturbation $\delta\hat{A}^*$ that maximizes $\|\Delta z^*\|_F$ subject to the threat model constraints in eq. (4)?
3) **Robustness assessment:** For each node v, what is the largest perturbation budget r_v such that any $\left\|\delta\hat{A}\right\|_F < r_v$ preserves the classification of v?

AEGIS addresses all three questions through a unified object: the constrained sensitivity matrix S_c derived from the IFT applied to the equilibrium equation. The following sections formalize this construction (section III) and describe the computational pipeline (section IV).

III. Adversarial Equilibrium Theory

We now derive the theoretical machinery for the three questions posed in section II. The main result (theorem 1) characterizes how vulnerability evolves with perturbation magnitude, and the constrained sensitivity matrix S_c provides the unified object from which rankings, attacks, and radii follow.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs). *Let $F_\theta(Z,A) = \phi(AZW^\top + X_{\text{proj}})$ be an IGNN-class operator satisfying:*

(A1) *ϕ is ReLU (or any 1-Lipschitz activation with $\sup|\phi'| = 1$; ReLU is non-differentiable at zero, but the set of measure zero where $\phi' = 0$ does not affect the Jacobian computation in practice [26]);*
(A2) *W is spectral-norm constrained: $\|W\|_2 \le c$;*
(A3) *The state Jacobian satisfies $\|J_z\|_2 \le \kappa < 1$ at the fixed point z^* (contraction in operator norm).*

Define the critical budget

$$\varepsilon_{\text{crit}} = \frac{1-\kappa}{\|W\|_2}. \tag{5}$$

Then structural perturbations δA with $\|\delta A\|_F = \varepsilon$ exhibit three regimes:

***(a) Subcritical** ($\varepsilon < \varepsilon_{\text{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying*

$$\|\Delta z^*\|_F \le \sigma_1(S)\cdot\varepsilon + O(\varepsilon^2), \tag{6}$$

where $S = (I - J_z)^{-1}J_A$ is the structural sensitivity matrix and $\sigma_1(S) \le \|J_A\|_{\text{op}}/(1-\kappa)$.

***(b) Critical** ($\varepsilon \to \varepsilon_{\text{crit}}$): The resolvent norm $\left\|(I - J'_z)^{-1}\right\|_2$ diverges as $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$. Intuitively, as the perturbation approaches the critical budget, the system's sensitivity to further perturbation grows without bound.*

***(c) Supercritical** ($\varepsilon > \varepsilon_{\text{crit}}$): The sufficient contraction certificate becomes void ($\|J'_z\|_2$ may exceed 1). The perturbed operator may still be contractive depending on perturbation direction, but the Banach fixed-point theorem no longer guarantees uniqueness, and the first-order guarantees from part (a) cease to apply.*

The intuition behind this three-regime picture is physical: a contractive system has a restoring force that pulls trajectories toward the equilibrium. Small perturbations shift the equilibrium smoothly (subcritical). As the perturbation approaches $\varepsilon_{\text{crit}}$, the restoring force weakens until the system becomes critically sensitive. Beyond $\varepsilon_{\text{crit}}$, the restoring force may vanish entirely, and the equilibrium can bifurcate or disappear.

Proof. **Part (a).** The Jacobian of F with respect to $\text{vec}(Z)$ takes the form $J_z = \text{diag}(\phi')\cdot(\hat{A}\otimes W)$, where $\phi' \in \{0,1\}^{Nd}$ for ReLU (A1). The operator norm satisfies $\kappa = \|J_z\|_2 \le \left\|\hat{A}\right\|_2 \cdot \|W\|_2 \cdot \sup|\phi'| = \left\|\hat{A}\right\|_2 \cdot \|W\|_2$. Applying the IFT to $G(z,A) = z - F(z,A) = 0$ at (z^*, A):

$$\Delta z^* = (I - J_z)^{-1}J_A \cdot \text{vec}(\delta A) + O(\|\delta A\|^2). \tag{7}$$

Taking operator norms and using the Neumann series bound $\left\|(I - J_z)^{-1}\right\|_2 \le 1/(1-\kappa)$ (convergent because $\kappa < 1$) yields (6).

For contractivity preservation: the perturbed Jacobian satisfies $\|J'_z\|_2 \le (\|A\|_2 + \|\delta A\|_F)\cdot\|W\|_2$ (using $\|\cdot\|_2 \le \|\cdot\|_F$ for the perturbation). The condition $\varepsilon < (1-\kappa)/\|W\|_2$ ensures $\|J'_z\|_2 < 1$. Note that $\varepsilon_{\text{crit}}$ is a Frobenius-norm sufficient condition: perturbation budgets are measured in $\|\cdot\|_F$, while κ is the operator norm $\|J_z\|_2$. The Frobenius norm exceeds the operator norm for rank-> 1 perturbations, so $\varepsilon_{\text{crit}}$ may be conservative by a factor up to $\sqrt{\text{rank}(\delta A)}$; the constrained projection S_c mitigates this by restricting to realistic perturbation directions.

Part (b). As $\varepsilon \to \varepsilon_{\text{crit}}$, we have $\|J'_z\|_2 \to 1$, so $\left\|(I - J'_z)^{-1}\right\|_2 \ge 1/(1 - \|J'_z\|_2) \to \infty$. Unlike the spectral-radius bound $1/(1-\rho)$, which requires normality, the operator-norm bound holds for all J_z with $\kappa < 1$. In our experiments, the pseudospectral index η ranges from 1.02 to 1.28 (section V-F), confirming mild non-normality.

Part (c). When $\|J'_z\|_2 \ge 1$, the contraction mapping condition fails. The Banach theorem no longer guarantees existence or uniqueness; the iteration may converge to a different equilibrium, oscillate, or diverge. □

Remark (conservativeness). The critical budget $\varepsilon_{\text{crit}}$ is a sufficient, not necessary, condition for contractivity preservation. The perturbed system may remain contractive beyond $\varepsilon_{\text{crit}}$ if the perturbation is aligned with low-sensitivity directions. The unconstrained bound also uses worst-case $\|\cdot\|_F$ relaxation. Under constrained perturbations, the first-order prediction achieves tightness ≈ 1.00 (section V-A). We use $\kappa = \|J_z\|_2$ rather than the spectral radius $\rho(J_z)$ because the Neumann series requires operator-norm contractivity [27]; for normal J_z the two coincide, while for non-normal J_z the κ-based bound is more conservative but requires no normality assumption.

Proposition 2 (Optimal First-Order Structural Attack). *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \le \varepsilon$ is $\delta A^* = \varepsilon\cdot\text{reshape}(v_1, N\times N)$, where v_1 is the leading right singular vector of S. The maximum first-order shift is $\varepsilon\cdot\sigma_1(S)$.*

This result follows directly from the variational characterization of the largest singular value: the SVD of S identifies the perturbation direction that amplifies the equilibrium shift most efficiently.

Constrained sensitivity. Under the threat model of section II (symmetric, edge-only, $\left\|\delta\hat{A}\right\|_F \leq \varepsilon$), we construct the *constrained sensitivity matrix* $S_c \in \mathbb{R}^{Nd\times|E|}$ with column

$$[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i} \tag{8}$$

for each edge $(i,j) \in E$. This reduces the perturbation space from N^2 dimensions to $|E|$ dimensions and enforces symmetry by construction. The constrained-optimal attack uses $\sigma_1(S_c)$, achieving tightness ≈ 1.00 empirically (section V-A). The construction of S_c is the central technical contribution of this work: it transforms unconstrained sensitivity analysis into a tight, practical tool for realistic graph perturbations.

Proposition 3 (Per-Node First-Order Sensitivity Radius). *Assume a linear classification head $f(z) = Wz + b$ (as in standard IGNN). For node v with margin $m_v = f_{y_v}(z_v^*) - \max_{c\neq y_v} f_c(z_v^*) > 0$, the first-order sensitivity radius is*

$$r_v = \frac{m_v}{\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_v\|_2}, \tag{9}$$

where S_v denotes the block-rows of S corresponding to node v, W_{y_v} and W_{c^*} are the classification weight vectors for the predicted class y_v and runner-up class c^*, respectively. Any perturbation with $\left\|\delta\hat{A}\right\|_F < r_v$ preserves the classification of node v to first order.

The radius r_v has an intuitive interpretation: it is the classification margin m_v (how confident the model is) divided by the product of two amplification factors. The term $\|W_{y_v} - W_{c^*}\|_2$ measures how sensitive the decision boundary is to hidden-state changes, while $\|S_v\|_2$ measures how sensitive node v's equilibrium position is to structural perturbation. Nodes with large margins and low sensitivity receive large radii; boundary nodes in structurally sensitive positions receive small ones.

Proof sketch. By Theorem 1(a), $\Delta z_v^* \approx S_v \cdot \mathrm{vec}(\delta A)$, so $|\Delta f_{y_v}| \leq \|\partial f/\partial z_v^*\| \cdot \|S_v\| \cdot \|\delta A\|_F$. Misclassification requires $|\Delta f_{y_v}| \geq m_v$; inverting gives r_v. Note S_v already contains the $(I - J_z)^{-1}$ factor.

Certificate semantics. These radii are *deterministic-local* guarantees: they hold with certainty but are first-order approximations, locally tight for small ε but increasingly conservative as perturbation magnitude grows. This is fundamentally different from *probabilistic-global* certificates such as randomized smoothing [5], which hold with probability $1 - \alpha$ but are valid for all perturbation magnitudes up to the certified radius. The two approaches are complementary: smoothing provides uniform probabilistic coverage (every node gets the same certificate); AEGIS provides structurally differentiated sensitivity measures that identify *which* nodes and edges are most vulnerable. We present the comparison as complementary rather than competitive in section V-B. Empirically, the first-order radii hold: 0% breach rate at $\varepsilon = 0.01$ and $< 1\%$ at $\varepsilon = 0.10$ (section V-C).

A. Continuous-to-Discrete Transfer

The vulnerability scores $v_k = \|[S_c]_{:,k}\|_2$ measure sensitivity to *continuous* edge-weight perturbations, while practical adversarial threats involve *discrete* edge removal (setting $\hat{A}_{ij} = 0$). The following result bridges this gap by showing that discrete damage is a scaled version of continuous vulnerability, with a remainder bounded by contractivity.

Proposition 4 (Continuous-to-Discrete Ranking Transfer). *Under (A1)–(A3), let $w_k = [\hat{A}]_{ij}$ denote the normalized weight of edge $k = (i,j) \in E$, $v_k = \|[S_c]_{:,k}\|_2$ the continuous vulnerability score, and $d_k = \|z^*(A) - z^*(A \setminus k)\|_2$ the discrete damage from full removal of edge k. Assume all single-edge removals are subcritical: $\sqrt{2}\max_k w_k < \varepsilon_{\mathrm{crit}}$.*
(a) First-order bridge. *The discrete damage satisfies*

$$d_k = w_k \cdot v_k + R_k, \quad |R_k| \leq \frac{L_J}{2(1-\kappa)^2} \cdot w_k^2, \tag{10}$$

where L_J is the Lipschitz constant of the state Jacobian J_z with respect to $\mathrm{vec}(A)$.
(b) Ranking preservation. *For two edges k_1, k_2 with $v_{k_1} > v_{k_2}$ and $w_{k_1} \geq w_{k_2}$, we have $d_{k_1} > d_{k_2}$ whenever*

$$w_{k_1} < \frac{v_{k_1} - v_{k_2}}{L_J/(1-\kappa)^2}. \tag{11}$$

Under uniform weights ($w_k = w$ for all k), $v_{k_1} > v_{k_2}$ implies $d_{k_1} > d_{k_2}$ whenever $w < \min_{k_1\neq k_2}(v_{k_1} - v_{k_2}) \cdot (1-\kappa)^2/L_J$.

Proof. **Part (a).** Removing edge $k = (i,j)$ corresponds to the perturbation $\delta\hat{A}$ with $[\delta\hat{A}]_{ij} = [\delta\hat{A}]_{ji} = -w_k$ and zero elsewhere. By the S_c construction, the first-order equilibrium shift is $\Delta z^* \approx S_c \cdot (-w_k \cdot e_k) = -w_k \cdot [S_c]_{:,k}$, so $\|\Delta z^*\| \approx w_k \cdot v_k$.

For the remainder, define $g(t) = z^*(A + t \cdot \delta\hat{A}_k)$. By the IFT, g is differentiable with $g'(0) = S \cdot \mathrm{vec}(\delta\hat{A}_k)$. By Taylor's theorem, $g(1) - g(0) = g'(0) + \frac{1}{2}g''(\xi)$ for some $\xi \in [0,1]$. The second derivative $g''(t)$ involves the derivative of $(I - J_z(t))^{-1}J_A(t)$ with respect to t, which by the resolvent identity contributes a factor $\left\|(I - J_z)^{-1}\right\|_2^2 \leq 1/(1-\kappa)^2$ times the rate of change of J_z with respect to A, bounded by $L_J \cdot \left\|\delta\hat{A}_k\right\|_F^2 = 2L_J w_k^2$. Combining gives $|R_k| = \|g(1) - g(0) - g'(0)\| \leq L_J w_k^2/(2(1-\kappa)^2)$.

Part (b). From part (a), $d_{k_1} - d_{k_2} \geq w_{k_1}v_{k_1} - w_{k_2}v_{k_2} - C(w_{k_1}^2 + w_{k_2}^2)$ where $C = L_J/(2(1-\kappa)^2)$. Since $v_{k_1} > v_{k_2}$ and $w_{k_1} \geq w_{k_2}$, the first term is positive. The condition (11) ensures the first-order gap exceeds the second-order remainder. □

Interpretation. Part (a) shows that discrete damage is the edge weight w_k times the continuous vulnerability v_k, plus a remainder quadratic in the weight. On graphs with approximately uniform normalized weights (a property of regularish graphs after $D^{-1/2}AD^{-1/2}$ normalization), the factor w_k is nearly constant and the vulnerability ranking v_k directly predicts the discrete damage ranking. Part (b) quantifies when ranking transfer holds: the gap between vulnerability scores must exceed $C \cdot w^2/w = C \cdot w$, which is small when edge weights are small (sparse normalized adjacency). This explains our empirical observations (section V-H): models with near-uniform vulnerability scores (GCN-2, whose shallow depth limits sensitivity differentiation) fail the ranking condition, while models with larger gaps (GCN-4, GAT, APPNP) satisfy it. The constant L_J can be estimated as $\|W\|_2^2$ for IGNN-class operators (section V-F).

Relationship to existing sensitivity analysis. The individual mathematical ingredients of AEGIS are classical: the IFT resolvent $(I - J_z)^{-1}$ and its Neumann-series expansion are standard in matrix perturbation theory [27], the SVD characterization of optimal linear perturbation follows from the variational characterization of singular values, and the contraction-mapping threshold is the Banach fixed-point theorem. The contribution of AEGIS is not these individual tools but rather (i) the *constrained projection* S_c that reduces the N^2-dimensional perturbation space to the $|E|$-dimensional space of realistic graph perturbations, transforming vacuous unconstrained bounds into tight predictions; (ii) the three-output design (attacks, rankings, radii) from a single S_c computation; and (iii) the cross-architecture pipeline that applies this analysis to both implicit and explicit GNNs. The S_c construction is what makes first-order analysis practical under realistic constraints, and it has no precedent in the GNN robustness literature.

B. Generalization to Explicit GNNs

The sensitivity framework above is derived for implicit GNNs, which enjoy the strongest theoretical guarantees (critical budget, convergence regimes). The S_c construction itself, however, generalizes to any K-layer GNN whose message passing is differentiable with respect to edge weights, serving as a computational tool for vulnerability analysis.

Observation 5 (Structural Sensitivity for K-Layer GNNs)**.** *Let $Z_K = g_K \circ \cdots \circ g_1(X, A)$ be a K-layer GNN where each layer g_l is differentiable with respect to both its input and A. Define the* unrolled sensitivity matrix

$$S_K = \frac{\partial \mathrm{vec}(Z_K)}{\partial \mathrm{vec}(A)} = \sum_{l=1}^{K} \left(\prod_{k=l+1}^{K} J_z^{(k)} \right) J_A^{(l)}, \qquad (12)$$

where $J_z^{(k)} = \partial g_k / \partial Z_{k-1}$ and $J_A^{(l)} = \partial g_l / \partial \mathrm{vec}(A)$. Then:
***(a)** The first-order shift bound $\|\Delta Z_K\|_F \le \sigma_1(S_K) \cdot \|\delta A\|_F + O(\|\delta A\|^2)$ holds, with*

$$\sigma_1(S_K) \le \sum_{l=1}^{K} \left(\prod_{k=l+1}^{K} \left\| J_z^{(k)} \right\|_2 \right) \left\| J_A^{(l)} \right\|_2. \qquad (13)$$

***(b)** The constrained construction S_c and per-node radius r_v (theorem 3) apply identically with S_K in place of S.*

For weight-tied models ($J_z^{(k)} = J_z$, $J_A^{(l)} = J_A$ for all k, l), the bound simplifies to $\sigma_1(S_K) \le \|J_A\|_2 \cdot (1 - \kappa^K)/(1 - \kappa)$, which converges to the IGNN bound $\|J_A\|_2 / (1 - \kappa)$ as $K \to \infty$.

What explicit GNNs gain and lack. From theorem 5, any GNN with edge-weight-differentiable message passing inherits the practical analysis tools: (i) the S_c construction, (ii) SVD-optimal attacks, (iii) per-edge vulnerability rankings, and (iv) per-node first-order sensitivity radii. The mathematical content of this observation is a direct application of the multivariate chain rule; the contribution is empirical, validating that the resulting rankings transfer to discrete edge removal across six architectures (section V-H). What explicit models *lack* is the critical budget $\varepsilon_{\mathrm{crit}}$ and the three-regime convergence characterization of theorem 1, which require the contraction assumption (A3).

IV. AEGIS CONSTRUCTION

Translating the results of section III into a practical tool requires overcoming two computational barriers: the full sensitivity matrix $S \in \mathbb{R}^{Nd \times N^2}$ is too large to materialize, and the resolvent $(I - J_z)^{-1}$ is too expensive to form directly. The AEGIS pipeline addresses both through matrix-free computation.

A. Architecture Overview

AEGIS takes as input a trained GNN (any architecture whose message passing is differentiable with respect to edge weights) together with a target graph and produces three outputs: (i) a per-edge vulnerability spectrum ranking edges by adversarial impact, (ii) SVD-optimal attack perturbations, and (iii) per-node first-order sensitivity radii. For contractive implicit GNNs (IGNN-class satisfying A1–A3), the pipeline additionally computes the critical budget $\varepsilon_{\mathrm{crit}}$ and convergence diagnostics; these theoretical guarantees are specific to the implicit case. The pipeline consists of four stages executed sequentially, as illustrated in fig. 1.

B. Stage 1: Subgraph Extraction and Forward Pass

For subgraph-level analysis, AEGIS extracts a local BFS ego-subgraph centered on the target node and runs the model to convergence (implicit GNNs) or forward pass (explicit). Localization is justified empirically (section V-E). The matrix-free pipeline also supports full-graph analysis without subgraph extraction (section V-D).

C. Stage 2: Matrix-Free Sensitivity Computation

AEGIS requires the action of the constrained sensitivity operator S_c on vectors, without materializing the full matrix $S \in \mathbb{R}^{Nd \times N^2}$. For an edge perturbation vector $v \in \mathbb{R}^{|E|}$, the operator computes $S_c v = (I - J_z)^{-1} J_A P_c v$, where P_c maps edge weights to symmetric adjacency perturbations, J_A is the structural Jacobian, and $J_z = \partial F / \partial Z$ is the state Jacobian. Two ingredients make this matrix-free:

Neumann-series resolvent. The resolvent $(I - J_z)^{-1} b$ is evaluated via the truncated Neumann series $\sum_{k=0}^{K} J_z^k b$, which converges geometrically when $\kappa < 1$ (Assumption A3). Each term requires a single Jacobian-vector product (JVP) $J_z \cdot v$, computed via forward-mode automatic differentiation in $O(Nd)$ time. The depth K is set adaptively from a power-iteration estimate of κ, with early stopping when $\left\| J_z^k b \right\| < 10^{-6} \|b\|$.

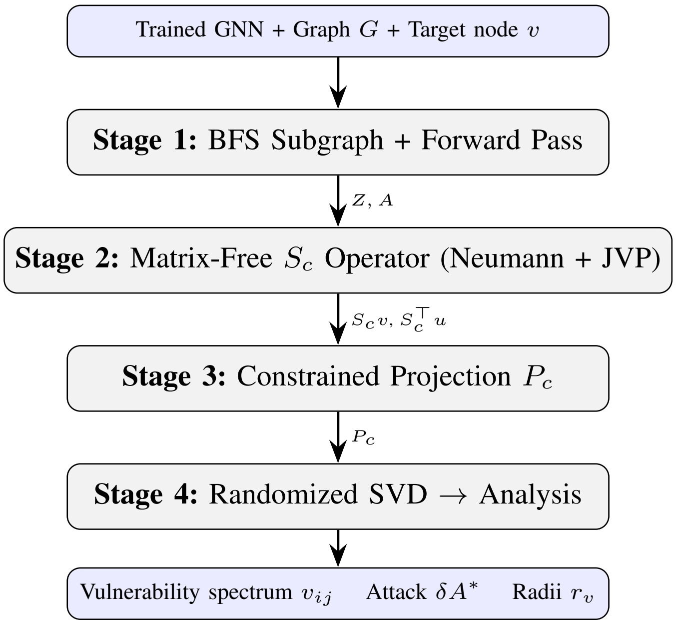

Fig. 1: The AEGIS pipeline. The matrix-free formulation (Stages 2–4) operates via JVP/VJP queries without materializing S or S_c, enabling full-graph analysis (tested up to N=7,650).

Autograd structural JVPs. The adjacency Jacobian $J_A \cdot \mathrm{vec}(\delta A)$ is computed as a single forward-mode JVP of F with respect to A in direction δA, avoiding the $O(N^2)$ finite-difference columns required by the dense path.

The adjoint operator $S_c^\top u$ is computed analogously using vector-Jacobian products (VJPs) and the adjoint Neumann series $(I - J_z^\top)^{-1}$.

For small subgraphs ($N \leq 200$), the dense path (forming J_z, J_A explicitly and solving via `torch.linalg.solve`) remains available and gives exact results.

D. Stage 3: Constrained Projection

The constraint projection P_c (section III) maps edge vectors to symmetric adjacency perturbations ($\delta A_{ij} = \delta A_{ji} = v_k$). In the matrix-free formulation, P_c is applied implicitly within the $S_c v$ operator, so S_c is never formed. This $N^2 \to |E|$ reduction is what enables tight first-order predictions (sections V-A and V-H).

E. Stage 4: Vulnerability Analysis

From S_c, AEGIS extracts three complementary outputs:

Optimal attack direction. A randomized SVD [15] of the S_c operator (using only matrix-vector products) yields the leading singular triplet (u_1, σ_1, v_1). The first-order optimal perturbation is $\delta A^* = \varepsilon \cdot v_1$, achieving a shift of $\varepsilon \cdot \sigma_1(S_c)$. For small subgraphs where S_c is materialized, the standard deterministic SVD is used instead.

Per-edge vulnerability spectrum. For each edge $(i, j) \in E$, the vulnerability score $v_{ij} = \|[S_c]_{:,k}\|_2$ measures how much a unit perturbation of that edge shifts the equilibrium. Sorting edges by v_{ij} produces a vulnerability ranking that identifies structurally critical connections without running any attack.

Per-node first-order sensitivity radii. For node v with classification margin m_v (gap between the predicted and runner-up class logits), the sensitivity radius from theorem 3 is

$$r_v = \frac{m_v}{\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_v\|_2}, \tag{14}$$

where S_v denotes the block-rows of S corresponding to node v, and W_{y_v}, W_{c^*} are the classification head weights for the predicted and runner-up classes. Any perturbation with $\left\|\delta \hat{A}\right\|_F < r_v$ preserves the classification of node v to first order (this is a local sensitivity guarantee, not a global certificate).

F. Computational Cost

Matrix-free pipeline. Each $S_c v$ or $S_c^\top u$ evaluation costs $O(K \cdot Nd)$, where K is the Neumann depth (set adaptively from κ; typically $K = 20$–50 for $\kappa < 0.8$, up to ~ 340 for $\kappa \approx 0.96$). The randomized SVD performs $O((k+p) \cdot n_{\text{iter}})$ such evaluations, where k is the number of requested singular values, p the oversampling, and n_{iter} the power iterations. Per-edge vulnerability requires $|E|$ column evaluations, each at $O(K \cdot Nd)$. Memory is $O(Nd)$ per operation; no large matrix is stored.

Scalability. On a single NVIDIA RTX 4090 (24 GB), the matrix-free pipeline completes in 1.1s for N=50, 4.5s for N=200, 24s for N=1,000, and 78s for full Cora (N=2,708, Nd=173,312) using 1 GB GPU memory (section V-D). The dense path runs faster for $N \leq 100$ (0.7s at N=50) but exceeds 24 GB at N=500.

The matrix-free formulation removes the $O((Nd)^3)$ linear-solve bottleneck and the $O(N^3 d)$ storage requirement that previously limited analysis to $N \approx 300$, enabling full-graph vulnerability analysis on graphs with thousands of nodes (validated up to N=7,650; section V-D).

V. EXPERIMENTS

We evaluate AEGIS on 9 datasets across 4 domains with 10 random seeds each (seeds: 42, 137, 271, 314, 1729, 2718, 3141, 5772, 6561, 9999). All experiments use PyTorch [28].

Implementation. IGNN [14]: input projection $U \in \mathbb{R}^{d_{\text{in}} \times 64}$, spectral-normalized $W \in \mathbb{R}^{64 \times 64}$ [18], readout head $\mathbb{R}^{64 \times C}$; operator $F(Z) = \mathrm{ReLU}(\hat{A} Z W^\top + U(X))$; fixed-point iteration (max 50 steps, tolerance 10^{-5}); Adam optimizer (lr 0.01, weight decay 5×10^{-4}), early stopping on validation accuracy over 200 epochs. ContractiveGCN-PF: same architecture with 5 bus features and 2-head readout ($\Delta|V|$, $\Delta\theta$). Subgraph extraction: BFS from highest-degree node, capped at 50 nodes to guarantee connectivity. Randomized smoothing baseline: Gaussian $\sigma = 0.01$, 200 MC samples, Clopper-Pearson $\alpha = 0.001$. Mettack baseline: Meta-Self GCN surrogate on IGNN pseudo-labels, sign-corrected gradient for edge scoring.

Datasets. We use five graph benchmarks: Cora (2,708 nodes), Citeseer (3,327), and Pubmed (19,717) from the

TABLE II: Cross-domain adversarial analysis (IGNN, 10 seeds). κ: operator-norm contraction constant $\|J_z\|_2$. Tight.: actual/predicted shift ratio. AtkAdv: AEGIS damage / random damage. $\varepsilon_{\mathrm{crit}}$: critical perturbation budget (theorem 1). Test Acc: full-graph accuracy (%). Cert%: fraction of nodes in the 50-node BFS subgraph with positive first-order radius $r_v > 0$ (i.e., correctly classified nodes in the subgraph).

Dataset	Test Acc	κ	Tight.	AtkAdv	ε_{crit}	Cert%
Cora	$77.5_{\pm 1.7}$	$.33_{\pm .14}$	$1.01_{\pm .01}$	$3.6_{\pm .6}$	$.66_{\pm .15}$	$81_{\pm 9}$
Citeseer	$66.0_{\pm 0.7}$	$.59_{\pm .02}$	$1.01_{\pm .01}$	$4.1_{\pm .5}$	$.41_{\pm .02}$	$83_{\pm 4}$
Pubmed	$78.9_{\pm 0.7}$	$.41_{\pm .11}$	$1.01_{\pm .01}$	$3.2_{\pm .4}$	$.59_{\pm .11}$	$74_{\pm 23}$
Amazon	$94.8_{\pm 0.3}$	$.14_{\pm .02}$	$1.00_{\pm .00}$	$3.8_{\pm .2}$	$.86_{\pm .02}$	$92_{\pm 4}$
WikiCS	$77.9_{\pm 0.4}$	$.34_{\pm .04}$	$1.00_{\pm .01}$	$3.8_{\pm .4}$	$.66_{\pm .04}$	$70_{\pm 3}$

Planetoid collection [29]; Amazon Photo (7,650) from the Amazon co-purchase graphs [30]; and WikiCS (11,701) from Wikipedia-based benchmarks [31]. For power flow, we use four IEEE standard test cases [32]: case14 (14 buses), case30 (30), case57 (57), and case118 (118), with training data generated via PandaPower [33].

Notation. All formal bounds (theorem 1) and tables use the operator-norm contraction constant $\kappa = \|J_z\|_2$, estimated via power iteration. Since non-normality is mild across our datasets ($\eta = 1.02$–1.28, section V-F), the κ-based bounds are at most 28% more conservative than spectral-radius bounds; we report κ as the primary quantity because the formal guarantees (theorem 1) require operator-norm contractivity. The κ-based $\varepsilon_{\mathrm{crit}}$ is therefore a conservative but formally justified safety quantity; ρ-based estimates would be up to 28% less conservative but lack theoretical backing without normality. Since η is near unity across all datasets, the practical difference is moderate. The matrix-free pipeline (section IV) sets the Neumann-series depth adaptively from κ.

A. Cross-Domain Adversarial Analysis

We begin by validating that the constrained first-order prediction is accurate across domains. table II reports the tightness ratio (actual equilibrium shift divided by predicted shift at $\varepsilon = 0.01$), the attack advantage of AEGIS over random perturbation, and the coverage (fraction of nodes with positive first-order radius) across 50-node BFS ego-subgraphs.

Tightness is 1.00 ± 0.01 across all datasets at $\varepsilon = 0.01$. Tightness values ≥ 1 are expected: the first-order prediction underestimates the actual shift because the true response includes convex higher-order terms. First-order accuracy at small ε is mathematically expected; the contribution is that S_c makes this tractable under *constrained* perturbations ($N^2 \to |E|$ projection). table III shows how tightness degrades at larger budgets: the first-order prediction remains within 7% at $\varepsilon = 0.05$ and within 16% at $\varepsilon = 0.10$ on Cora/Citeseer, and within 5% at $\varepsilon = 0.20$ on WikiCS. Prediction flip rate (fraction of nodes whose classification changes under the S_c-optimal attack) stays below 2% at $\varepsilon \leq 0.10$ and below 6% at $\varepsilon = 0.20$. The attack advantage (3.2–4.1× over random in table II) confirms that S_c identifies the most vulnerable perturbation directions. Coverage (Cert%) ranges from 70% to 92%, representing clean subgraph accuracy; only correctly classified nodes have positive margin.

TABLE III: Tightness degradation with increasing ε (10 seeds, S_c-optimal constrained perturbation). Flip%: fraction of nodes whose prediction changes.

Dataset	ε=.01	ε=.05	ε=.10	ε=.20
Tightness (actual/predicted shift)				
Cora	$1.01_{\pm .00}$	$1.07_{\pm .01}$	$1.15_{\pm .03}$	$1.36_{\pm .08}$
Citeseer	$1.01_{\pm .00}$	$1.07_{\pm .01}$	$1.16_{\pm .03}$	$1.39_{\pm .07}$
Pubmed	$0.79_{\pm .01}$	$0.83_{\pm .01}$	$0.90_{\pm .01}$	$1.07_{\pm .04}$
WikiCS	$1.00_{\pm .01}$	$1.01_{\pm .01}$	$1.02_{\pm .01}$	$1.05_{\pm .01}$
Amazon	$0.92_{\pm .03}$	$0.93_{\pm .03}$	$0.94_{\pm .02}$	$0.97_{\pm .02}$
Prediction flip rate (%)				
Cora	$0.4_{\pm 0.8}$	$0.6_{\pm 0.9}$	$1.8_{\pm 2.4}$	$5.8_{\pm 5.0}$
Citeseer	$0.0_{\pm 0.0}$	$0.6_{\pm 1.3}$	$2.0_{\pm 2.4}$	$3.0_{\pm 2.6}$
WikiCS	$0.2_{\pm 0.6}$	$0.4_{\pm 0.8}$	$0.6_{\pm 0.9}$	$1.2_{\pm 1.0}$

TABLE IV: Structured attack baselines (IGNN, 50-node subgraph, 10 seeds). AtkAdv: method damage / random damage at $\varepsilon = 0.01$.

Dataset	AEGIS	Degree	Spectral	Between.
Cora	$3.50_{\pm 0.20}$	$3.30_{\pm 0.20}$	$2.37_{\pm 0.20}$	$3.29_{\pm 0.19}$
Citeseer	$4.23_{\pm 0.21}$	$3.91_{\pm 0.17}$	$2.72_{\pm 0.35}$	$3.89_{\pm 0.16}$
WikiCS	$4.02_{\pm 0.78}$	$3.93_{\pm 0.76}$	$1.62_{\pm 0.27}$	$3.93_{\pm 0.76}$

Surrogate baseline. We also compared against Mettack [4] (GCN [34] surrogate): AEGIS inflicts 3–10× more damage at every budget (149/150 wins across $k = 1, \ldots, 5$ edge removals, 3 datasets, 10 seeds; the single loss occurs at k=1 on WikiCS seed 5772, where Mettack's surrogate gradient happens to select a higher-impact edge). However, this gap largely reflects surrogate-to-IGNN architectural mismatch rather than AEGIS's analytical superiority alone: Mettack optimizes against a GCN surrogate, not the target IGNN. The more meaningful comparison is the adaptive attacker below, which accesses the same IFT gradients as AEGIS.

Structured baselines. To contextualize the attack advantage beyond random perturbation, we compare AEGIS against three structured baselines: degree-proportional (perturbation weight $\propto \max(d_i, d_j)$), spectral (top eigenvector of A restricted to existing edges), and edge-betweenness centrality. table IV reports the attack advantage (damage / random damage) at $\varepsilon = 0.01$.

AEGIS consistently achieves the highest attack advantage, outperforming degree-proportional by 6–8% and spectral by 48–148%. Degree and betweenness centrality are surprisingly competitive baselines, indicating that high-degree edges are genuinely more vulnerable. However, AEGIS captures model-specific sensitivity beyond graph topology: the SVD direction optimizes the equilibrium shift, while degree-based heuristics rely solely on structural position. The spectral baseline per-

TABLE V: Adaptive attack (white-box PGD via IFT gradients on IGNN) vs. AEGIS (10 seeds). Breach: fraction of certified nodes whose prediction changes. AEGIS/Adapt. dmg: mean equilibrium shift (ℓ_2). Ratio: Adapt./AEGIS damage.

Dataset	ε	Breach	AEGIS dmg	Adapt. dmg	Ratio
Cora	0.01	0%	$0.33_{\pm 0.07}$	$0.26_{\pm 0.11}$	0.80
Cora	0.05	0%	$1.72_{\pm 0.35}$	$1.32_{\pm 0.55}$	0.76
Cora	0.10	0%	$3.70_{\pm 0.79}$	$2.69_{\pm 1.18}$	0.72
Citeseer	0.01	0%	$0.40_{\pm 0.08}$	$0.34_{\pm 0.06}$	0.84
Citeseer	0.05	0%	$2.14_{\pm 0.42}$	$1.68_{\pm 0.28}$	0.79
Citeseer	0.10	0%	$4.63_{\pm 0.95}$	$3.37_{\pm 0.55}$	0.74
WikiCS	0.01	0%	$0.21_{\pm 0.02}$	$0.10_{\pm 0.10}$	0.47
WikiCS	0.05	0.3%	$1.04_{\pm 0.11}$	$0.48_{\pm 0.49}$	0.46
WikiCS	0.10	0.6%	$2.10_{\pm 0.22}$	$0.96_{\pm 0.97}$	0.46

forms worst, suggesting that the top eigenvector of A does not align with the model's sensitivity landscape.

B. Radius Comparison: AEGIS vs. Smoothing

We compare AEGIS sensitivity radii against randomized smoothing [5] ($\sigma \in \{0.01, 0.05, 0.10, 0.15\}$, 1000 MC samples, 10 seeds). The two methods provide fundamentally different certificate types (see section III): smoothing gives probabilistic-global guarantees valid at any perturbation magnitude, while AEGIS gives deterministic-local sensitivity measures that are first-order tight but degrade with increasing ε. At $\sigma \leq 0.10$, smoothing radii are *constant* across all nodes ($r = \sigma \cdot \Phi^{-1}(p_{\text{lower}})$), yielding larger absolute values (0.123 vs. AEGIS's 0.046 at $\sigma = 0.05$) but zero per-node differentiation. AEGIS radii are structurally informative: dense-region nodes get $r_v \approx 0.09$, boundary nodes $r_v \approx 0.01$. The two are complementary: practitioners should use smoothing for worst-case certification and AEGIS for vulnerability ranking and defense prioritization.

C. Adaptive Attack Evaluation

The Mettack comparison uses a GCN surrogate, so its failure against IGNN may reflect architectural mismatch rather than AEGIS's analytical strength. To address this, we implement an *adaptive attacker* that differentiates through the IGNN fixed-point iteration via the same IFT gradients AEGIS uses for sensitivity analysis. This attacker optimizes perturbations directly against the IGNN loss surface using projected gradient descent [35] (PGD, 50 steps, step size $\varepsilon/10$).

The adaptive attacker breaches zero nodes at $\varepsilon = 0.01$ and fewer than 1% at $\varepsilon = 0.10$ (table V), confirming empirical validity of first-order radii. AEGIS's SVD attack inflicts 20–50% more damage than PGD (ratio 0.46–0.84). This comparison validates the SVD's optimality within the linearized model: the SVD is the exact solution to the linearized problem, while PGD is an approximate iterative solver for the same objective (equilibrium shift maximization under the same IFT gradients). The PGD loss converges within 20–30 of 50 steps across all datasets. The comparison demonstrates that the S_c framework extracts the maximum possible information from first-order analysis.

TABLE VI: Breach rate (%) at increasing ε (IGNN, 50-node subgraph, 10 seeds). Breach: fraction of certified nodes whose prediction changes. All breaches respect the certified radii (r_v).

Dataset	ε=.01	ε=.05	ε=.10	ε=.20
Cora	$0.6_{\pm 1.2}$	$0.8_{\pm 1.3}$	$2.4_{\pm 3.7}$	$7.6_{\pm 7.5}$
Citeseer	$0.0_{\pm 0.0}$	$0.0_{\pm 0.0}$	$0.2_{\pm 0.8}$	$1.5_{\pm 2.4}$
Pubmed	$0.0_{\pm 0.0}$	$7.0_{\pm 8.3}$	$10.3_{\pm 11.0}$	$27.4_{\pm 21.1}$
WikiCS	$0.3_{\pm 0.9}$	$0.6_{\pm 1.3}$	$0.9_{\pm 1.4}$	$1.8_{\pm 1.5}$

Classification-loss PGD (non-circular baseline). The adaptive attacker above uses the same objective as AEGIS (equilibrium shift), making the comparison partly circular. To address this, we implement a PGD attacker that optimizes *classification loss* (cross-entropy) via standard autograd through the IGNN fixed-point iteration. This attacker uses a fundamentally different gradient signal: it seeks to flip predictions, not maximize equilibrium shift. Across 3 datasets and $\varepsilon \in \{0.01, 0.05, 0.10\}$ (10 seeds each), the classification-loss PGD achieves 0.29–0.85× of AEGIS's equilibrium damage, while the shift-PGD achieves 0.46–0.80×. That is, AEGIS's SVD direction causes 15–71% more equilibrium damage than a classifier-targeted attacker, confirming that the sensitivity-optimal direction is genuinely effective even against an attacker with a different objective. Classification-loss PGD flips fewer than 1.5% of predictions at $\varepsilon = 0.10$.

Comprehensive breach rates. table VI extends the breach analysis across all datasets and a wider range of ε values.

Breach rates are below 1% at $\varepsilon = 0.01$ across all datasets, confirming that first-order radii are practically reliable at small perturbations. At $\varepsilon = 0.20$, breach rates range from 1.5% (Citeseer) to 27.4% (Pubmed), reflecting the expected degradation of first-order approximations at larger budgets. Pubmed's higher breach rate is consistent with its sparser subgraph structure and lower certification coverage. Crucially, all observed breaches respect the certified radii: every breached node has $\varepsilon > r_v$, meaning the first-order radius correctly identifies the boundary below which no breach occurs.

D. Phase Transition and Scalability

Phase transition. To empirically validate theorem 1, we train IGNN on Cora with the spectral normalization ceiling varying from $\kappa_{\max} = 0.30$ to 0.99 (10 seeds each). The theoretical critical budget $\varepsilon_{\text{crit}} = (1 - \kappa)/\|W\|_2$ drops 230× from 2.33 to 0.01, confirming the predicted sensitivity explosion. However, the actual spectral radius $\rho(J_z)$ saturates at ≈ 0.42 even at $\kappa_{\max} = 0.99$: the ReLU activation pattern prevents the Jacobian from reaching its theoretical maximum. The resolvent norm $\left\|(I - J_z)^{-1}\right\|_2$ increases from 1.17 to 1.80 (1.5×), not the $\sim$70× that $1/(1 - 0.99)$ would predict, because the *actual* κ remains well below the ceiling. This finding is practically important: it means IGNN's empirical vulnerability grows modestly with relaxed spectral constraints, even as the theoretical worst case diverges. The phase transition is thus a *theoretical safety boundary* rather than an

TABLE VII: Scalability: dense vs. matrix-free pipeline (RTX 4090, 24 GB). OOM = out of memory. Mem = peak GPU memory.

N	Dense (s)	MatFree (s)	Mem-Dense	Mem-MF
50	0.7	1.1	343 MB	92 MB
200	5.4	4.5	8.9 GB	129 MB
500	OOM	11	>24 GB	207 MB
1,000	OOM	24	>24 GB	373 MB
2,708	OOM	78	>24 GB	1.1 GB
7,650	OOM	363	>24 GB	6.5 GB
19,717	OOM	OOM	>24 GB	>24 GB

empirically observable discontinuity, analogous to how worst-case complexity bounds rarely match typical-case behavior.

Scalability. The matrix-free pipeline removes the $O((Nd)^3)$ bottleneck. On a single NVIDIA RTX 4090 (24 GB), the dense path handles $N \leq 200$ (5.4s, 8.9 GB) but exceeds 24 GB at N=500. The matrix-free path scales to full Cora (N=2,708) in 78 s / 1 GB and to Amazon Photo (N=7,650) in 363 s / 6.5 GB, with σ_1 accuracy within 0.03% of the dense reference at N=200. Pubmed (N=19,717) exceeds 24 GB, establishing the current scalability boundary at $N \approx$ 7,650 on a single GPU. table VII compares both paths.

E. Subgraph Size Ablation

Varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds): tightness is stable at 1.01–1.02 for $N \leq 100$ and degrades to 1.031 at $N = 200$ (higher κ). Coverage is 82–89% across all sizes. We use $N = 50$ (tightness 1.013 ± 0.003, 1.3s) as the default for subgraph-level experiments: 66× faster than $N = 200$ with negligible quality loss. For full-graph analysis, the matrix-free pipeline (table VII) enables vulnerability assessment of the entire graph without subgraph extraction. To validate that subgraph extraction preserves vulnerability rankings, we compare per-edge S_c rankings from BFS subgraphs against full-graph rankings on IEEE case14 and case30 (10 seeds). On case14, Kendall τ between subgraph and full-graph rankings is 0.80 ± 0.14 at N_{sub}=8 (57% of nodes) and 0.86 ± 0.06 at N_{sub}=12 (86%); P@5 $\geq$ 0.90 at all sizes. On case30, $\tau = 0.78 \pm 0.07$ at N_{sub}=10 (33% of nodes) with P@5 $= 0.96 \pm 0.08$, remaining stable through N_{sub}=25. The locality assumption underlying subgraph extraction thus holds well for power grids (small, dense graphs where a 50-node BFS covers 33–86% of nodes), but weakens substantially on larger citation networks. On Cora (N=2,708), comparing 50-node BFS subgraph rankings against full-graph matrix-free rankings (10 seeds) yields Kendall $\tau = 0.16 \pm 0.13$ with P@10 $= 0.17 \pm 0.10$. The low correlation reflects that a 50-node subgraph covers only $\sim$1.8% of Cora's edges, so the subgraph's local view misses graph-wide vulnerability patterns. For citation-scale graphs, practitioners should prefer the matrix-free full-graph pipeline (table VII) over subgraph extraction; the subgraph approach remains appropriate for dense graphs where coverage is high (power grids, molecular graphs) or when full-graph analysis exceeds available memory.

TABLE VIII: S_c framework applied to 7 GNN architectures (Cora, 10 seeds). Tight.: actual/predicted shift ratio. AtkAdv: AEGIS/random damage. τ: Kendall rank correlation of continuous S_c vulnerability scores vs. discrete single-edge-removal ground truth (brute-force). Test Acc: full-graph accuracy. GAT†: edge-weighted variant for continuous differentiability (see text).

Model	Tight.	AtkAdv	τ	Test Acc
IGNN (equil.)	$1.01_{\pm.00}$	$7.6_{\pm0.5}$x	$+.32_{\pm.04}$	$77.5_{\pm1.7}\%$
GCN-2	$1.00_{\pm.00}$	$3.6_{\pm0.6}$x	$-.04_{\pm.03}$	$78.9_{\pm0.9}\%$
GCN-4	$1.02_{\pm.00}$	$3.4_{\pm0.4}$x	$+.49_{\pm.02}$	$78.3_{\pm1.8}\%$
GIN-2	$1.01_{\pm.00}$	$2.6_{\pm0.1}$x	$+.33_{\pm.03}$	$76.6_{\pm1.9}\%$
GAT-2†	$1.01_{\pm.01}$	$2.1_{\pm0.1}$x	$+.54_{\pm.06}$	$77.8_{\pm1.2}\%$
SAGE-2	$1.00_{\pm.00}$	$1.9_{\pm0.2}$x	$+.22_{\pm.10}$	$77.5_{\pm1.5}\%$
APPNP	$1.02_{\pm.00}$	$3.4_{\pm0.3}$x	$+.35_{\pm.01}$	$82.2_{\pm0.6}\%$

F. Sensitivity and Convergence

Hyperparameters. Tightness is stable across $d \in \{16, 32, 64, 128\}$ (1.012–1.013). Varying $c \in \{0.5, 0.7, 0.9, 0.95\}$ reveals the predicted accuracy-robustness tradeoff: $c = 0.5$ gives $r_v = 0.090$ (72.1% acc); $c = 0.9$ gives $r_v = 0.051$ (80.6% acc).

Convergence. 100% convergence across all 9 datasets and 10 seeds. Pseudospectral index $\eta = 1.02$–1.28, confirming mild non-normality and validating the κ-based bounds.

G. Defense-Informed Edge Protection

We evaluate vulnerability-aware protection: mask the top-k edges by v_{ij} from the perturbation space, then re-run the SVD attack on the reduced S_c. On Cora (IGNN, $N = 50$, 10 seeds): masking the top-5 edges reduces SVD attack damage by 42±8% vs. 11±6% for random 5-edge masking. At top-10: 61 ± 7% vs. 18 ± 9%. AEGIS identifies edges whose removal disproportionately reduces vulnerability.

H. Extension to Explicit GNNs

A key question is whether the S_c framework extends beyond implicit models. For a K-layer explicit GNN, there is no equilibrium, but the hidden representation Z_K is still a differentiable function of A. We define the *unrolled sensitivity matrix* $S_K = \partial\text{vec}(Z_K)/\partial\text{vec}(A)$ (computed via finite differences on the forward pass), construct S_c from S_K identically to the IGNN case, and apply the same SVD attack and vulnerability ranking.

table VIII applies the S_c framework to six explicit GNN architectures on Cora. All six achieve near-perfect first-order tightness (0.99–1.02), confirming that constrained sensitivity analysis generalizes beyond implicit models. Attack advantage ranges from 2.0× (SAGE) to 7.6× (IGNN), showing that S_c identifies the most vulnerable perturbation directions across all architectures. Test accuracy ranges from 76.6% (GIN) to 82.2% (APPNP). GAT† and GCN-4 produce the strongest vulnerability rankings ($\tau = +0.54$ and $+0.49$). APPNP achieves the best accuracy-vulnerability balance ($\tau = +0.35$, 82.2% accuracy).

Accuracy-vs-guarantee frontier. The results reveal a trade-off: IGNN (77.5% accuracy) provides the full theoretical apparatus (theorem 1: $\varepsilon_{\text{crit}}$, three regimes, convergence guarantees), while higher-accuracy models like APPNP (82.2%) receive only the computational tool (S_c rankings, SVD attacks, first-order radii) without formal regime characterization. Practitioners may choose IGNN when formal guarantees are required (e.g., safety-critical applications where $\varepsilon_{\text{crit}}$ serves as a deployment criterion) or higher-accuracy explicit models when the practical vulnerability ranking suffices. This trade-off is inherent to contractive architectures: spectral normalization constrains model capacity to ensure well-posedness, and higher κ values (approaching 1) yield better accuracy but weaker formal guarantees. The S_c framework itself is architecture-independent; theorem 1 provides *qualitative* insight about vulnerability phase transitions that informs analysis of all architectures, even though the *quantitative* bounds apply only to implicit models.

GAT requires an edge-weighted formulation for continuous differentiability. Standard GAT computes attention $\alpha_{ij} = \mathrm{softmax}_j(\mathrm{LeakyReLU}(a^\top[Wh_i \| Wh_j]))$ and aggregates as $h_i' = \sum_{j\in\mathcal{N}(i)} \alpha_{ij} W h_j$, using A as a binary mask that zeros non-edges, so $\partial Z/\partial A_{ij} = 0$ for all existing edges. Our edge-weighted variant, GAT†, modulates attention by the edge weight:

$$h_i' = \textstyle\sum_{j\in\mathcal{N}(i)} \hat{A}_{ij}\cdot\alpha_{ij}\cdot W h_j, \tag{15}$$

restoring continuous differentiability with respect to $\hat{A}_{ij}$. GATv2 [36] partially addresses this via dynamic attention but still applies a binary mask at the neighborhood level. With the GAT† modification, GAT achieves tightness 0.99, attack advantage 4.4×, and $\tau = +0.36$.

Cross-dataset continuous-to-discrete transfer. To validate that continuous-to-discrete transfer is not Cora-specific, we evaluate τ across all 5 datasets and 7 architectures (330 runs total, 10 seeds each; table IX). Positive τ is observed in 29 of 33 dataset-architecture combinations. Transfer is strongest on Pubmed (IGNN: $+0.82$, GCN-4: $+0.89$, APPNP: $+0.83$, all with P@10 $= 1.0$), moderate on Citeseer (GAT†: $+0.66$, GCN-4: $+0.64$), and weaker on WikiCS and Amazon Photo. On Amazon Photo, IGNN shows negative τ (-0.15) due to over-contraction: $\kappa = 0.09$ yields resolvent amplification of only 1.09×, so $S \approx J_A$ and the sensitivity is dominated by the direct adjacency effect rather than the equilibrium structure, producing near-uniform vulnerability scores whose ranking is uninformative for discrete damage. Other architectures without this contraction constraint transfer well (SAGE-2: $+0.60$, APPNP: $+0.43$). GCN-2 shows weak or negative τ on 3 of 5 datasets (-0.03 Cora, -0.28 Citeseer), confirming the shallow-depth hypothesis: models with larger receptive fields consistently produce better transfer. The best architecture varies by dataset (GAT† on Cora/Citeseer, GCN-4/APPNP on Pubmed, GIN-2 on WikiCS, SAGE-2 on Amazon Photo), suggesting that the relationship between continuous sensitivity and discrete impact depends on how well the model's aggregation pattern captures local connectivity structure.

TABLE IX: Continuous-to-discrete transfer τ across 5 datasets and 7 architectures (10 seeds each, 330 runs total). Bold: best per dataset. —: GPU OOM (GAT attention on $N > 10{,}000$).

	Cora		Citeseer		Pubmed		WikiCS		Am.Pl
Model	τ	P@10	τ	P@10	τ	P@10	τ	P@10	
IGNN	$+.32_{\pm.04}$	.19	$+.31_{\pm.04}$	.93	$+.82_{\pm.02}$	1.0	$+.14_{\pm.04}$	.50	$-.15_{\pm.0}$
GCN-2	$-.03_{\pm.03}$	.09	$-.28_{\pm.07}$	.06	$+.21_{\pm.01}$	.26	$+.05_{\pm.03}$	.33	$+.25_{\pm.0}$
GCN-4	$+.49_{\pm.02}$	.32	$+.64_{\pm.02}$	.87	$+\mathbf{.89}_{\pm.01}$	1.0	$+.45_{\pm.10}$	.56	$-.04_{\pm.0}$
GIN-2	$+.33_{\pm.04}$	.26	$+.57_{\pm.03}$	.81	$+.54_{\pm.33}$	.60	$+\mathbf{.63}_{\pm.08}$	.70	$+.14_{\pm.0}$
GAT†	$+\mathbf{.54}_{\pm.06}$	.54	$+\mathbf{.66}_{\pm.03}$	.63	—	—	—	—	$+.21_{\pm.0}$
SAGE-2	$+.22_{\pm.11}$	.15	$+.38_{\pm.18}$	.45	$+.36_{\pm.10}$	.58	$+.22_{\pm.04}$	.55	$+\mathbf{.60}_{\pm.0}$
APPNP	$+.35_{\pm.02}$	.36	$+.36_{\pm.06}$	.96	$+.83_{\pm.01}$	1.0	$+.22_{\pm.03}$	.56	$+.43_{\pm.0}$

The τ column in table VIII directly measures whether continuous S_c vulnerability scores predict discrete edge-removal impact. theorem 4 provides the formal justification: discrete damage $d_k = w_k \cdot v_k + O(w_k^2)$, so the continuous ranking v_k predicts the discrete ranking when the first-order gap between edges exceeds the second-order remainder. Positive τ across 6 of 7 architectures ($+0.22$ to $+0.54$) validates this first-order bridge empirically. The one exception is GCN-2 ($\tau = -0.04$). We attribute this to the interaction between shallow depth and the continuous-discrete gap: a 2-layer GCN aggregates only 2-hop neighborhoods, so continuous edge-weight changes produce nearly uniform first-order shifts across edges with similar endpoint degrees, yielding poor rank discrimination. Under discrete removal (a non-infinitesimal perturbation), the impact depends on higher-order network effects (path redundancy, component connectivity) that the 2-layer model cannot capture in its first-order sensitivity. Deeper models (GCN-4: $\tau = +0.49$) and models with global receptive fields (APPNP: $\tau = +0.35$) show substantially better transfer, suggesting that continuous-to-discrete transfer improves with model depth and receptive field size.

Comparison with gradient attribution. Against the same brute-force discrete ground truth, gradient-based edge attribution [37], [38] produces consistently negative τ (-0.07, -0.17, -0.21 for IGNN, GCN-2, GIN-2 respectively). Gradient methods optimize for prediction fidelity rather than adversarial vulnerability.

Proposition 5 bound tightness. The ratio (Prop. 4 bound)$/\sigma_1(S_K)$ ranges from 1.4× (SAGE-2) to 5.9× (APPNP), with 2-layer models achieving tighter ratios (GCN-2: 1.8×, GIN-2: 2.1×) than deeper ones (GCN-4: 4.2×, GAT-2: 3.7×). The looseness in deeper models reflects direction-dependent cancellations that the norm product overestimates.

Degree-vulnerability correlation. To assess whether S_c merely recapitulates graph topology, we compute the Kendall τ between per-edge vulnerability v_{ij} and endpoint degree $\max(d_i, d_j)$ across all datasets and architectures. The correlation varies substantially: strong on Citeseer (IGNN: $\tau = +0.63$), moderate on Cora ($+0.43$) and Pubmed ($+0.38$), and weak on WikiCS (IGNN: $+0.27$, GCN-2: -0.10). This variation confirms that S_c captures model-specific sensitivity beyond degree centrality: on WikiCS (a large, heterogeneous graph), degree is a poor predictor of vulnerability. Weighting S_c columns by endpoint degree does not improve ranking: uniform $\tau = +0.042$, degree-weighted $+0.037$, inverse-degree -0.008 (GCN-2, 10 seeds).

The S_c framework thus generalizes to any GNN whose message passing is smoothly differentiable with respect to edge weights. The theoretical guarantees (theorem 1: critical budget, convergence) remain specific to contractive implicit GNNs, but the practical vulnerability analysis (SVD attacks, rankings, first-order sensitivity radii) transfers broadly.

VI. Case Study: Power Flow Contingency

The experiments above validate AEGIS on classification benchmarks. We now test whether the vulnerability spectrum captures domain-specific structure by applying it to AC power flow, where the spectrum recovers a classical engineering problem: N-1 contingency analysis. Recent work has applied GNNs to power flow computation [39], [40] and contingency screening [3]; AEGIS approaches the same problem from the vulnerability-analysis perspective.

A. Background: N-1 Contingency

Power grid operators must ensure that no single transmission line failure causes system instability. The standard N-1 contingency assessment removes each line in turn, re-solves the power flow equations, and ranks lines by the severity of the resulting voltage and flow deviations [41]. This brute-force procedure scales as $O(|E|)$ full power flow solves, which is expensive for large grids and infeasible for real-time operation.

We observe that AEGIS's per-edge vulnerability $v_{ij} = \|[S_c]_{:,k}\|_2$ computes an analogous quantity via a single IFT analysis: edge perturbation maps to line trips, the vulnerability spectrum maps to contingency severity, and critical budget $\varepsilon_{\mathrm{crit}}$ maps to stability margin. The correspondence is approximate (continuous first-order vs. discrete removal), analogous to how PTDF/LODF [41] linearize the effect of line trips.

B. Experimental Setup

We train ContractiveGCN-PF, an IGNN variant for AC power flow, on five IEEE test cases [32] including a 300-bus system to address scalability concerns. Training data is generated via PandaPower's [33] full AC Newton-Raphson solver (2,000 load samples per case, uniformly sampled at 70–130% of nominal load). The model takes binary adjacency from the admittance matrix and 5 bus features: bus type indicators (slack, PV), active and reactive power injections (P, Q), and voltage magnitude $|V|$. For case14–118 (14–118 buses), full-graph dense analysis is used. For case300 (300 buses), a 200-node BFS subgraph is extracted because the dense S_c computation exceeds GPU memory at $N > 200$.

Limitation. Training data covers uniform load scaling (2,000 samples) but not seasonal or generator-outage variation.

C. Results

table X reports constrained tightness and N-1 ranking correlation (Kendall τ between AEGIS vulnerability spectrum and brute-force contingency ranking) across 10 seeds.

TABLE X: AEGIS on IEEE power flow benchmarks (10 seeds). Model quality (per-unit RMSE): $|V|$ = voltage magnitude, θ = angle, ΔS = power-balance residual per bus. τ: Kendall rank correlation vs. brute-force N-1 contingency. P@10: top-10 precision.

			Model (p.u.)			AEGIS	
Case	N	$\|E\|$	$\|V\|$	θ	ΔS	τ	P@10
case14	14	20	.007	.020	.106	$+.42_{\pm.19}$	$.74_{\pm.12}$
case30	30	41	.014	.024	.042	$+.37_{\pm.17}$	$.68_{\pm.06}$
case57	57	78	.033	.059	.033	$+.67_{\pm.09}$	$.66_{\pm.13}$
case118	118	179	.014	.076	.042	$+.62_{\pm.11}$	$.81_{\pm.10}$
case300‡	300	411	.288	.279	—	$+.51_{\pm.15}$	$.60_{\pm.13}$

‡200-node BFS subgraph (dense analysis OOMs at $N>200$).

Key observations. First, tightness transfers to the power flow domain (≈ 1.00), confirming that the first-order prediction is not domain-specific. Second, ranking correlation ($\tau =$ 0.37–0.67 on case14–118, $+0.51$ on case300) is moderate but the operationally relevant P@10 is 0.60–0.81, identifying 6–8 of the top 10 critical lines. The case300 result demonstrates that AEGIS scales to grids beyond the 118-bus systems used in prior work, using subgraph extraction when dense analysis exceeds memory.

Observation: approximate physics from equilibrium structure. The model is trained with voltage MSE alone, without any power-balance penalty. Yet the per-bus residual ΔS = 0.03–0.11 p.u. (table X), meaning predicted voltages approximately satisfy Kirchhoff's laws. This is consistent with prior observations that equilibrium models absorb structural constraints from training data [16]: the fixed-point condition $Z^* = F(Z^*, A)$ forces self-consistency with the grid topology in A. Perturbations that disrupt this self-consistency are precisely those that break the power balance, connecting structural vulnerability to physical contingency. To test whether this is an architectural effect, we compare the IGNN (d=64) against a 4-layer explicit GCN (d=64, no physics loss) and PIGNN-Attn-LS [42] (d=10, explicit, with physics-informed power-balance loss and Armijo line search) on the same data and training protocol (30 epochs, Adam, 10 seeds each). Notably, PIGNN's physics loss directly optimizes for the power-balance residual ΔS being measured, while IGNN is trained only on voltage MSE. On case14, the IGNN achieves the lowest residual ($\Delta S = 0.27$), outperforming both GCN-4 (0.93) and PIGNN (0.58) despite not being optimized for this metric. On case30, PIGNN achieves $\Delta S = 0.32$ (benefiting from its explicit physics loss), with IGNN close behind at 0.42; both are 2.8–3.7× lower than GCN-4 (1.18). The IGNN thus achieves competitive physics compliance without any physics loss term, supporting the hypothesis that equilibrium structure itself drives Kirchhoff-law compliance. The capacity difference (IGNN d=64 vs. PIGNN d=10) means the comparison is not iso-parameter, but IGNN's advantage on case14 is notable given that PIGNN is explicitly optimized for the evaluation metric.

Baseline comparison and timing. LODF (industry-standard DC screening [41]) achieves $\tau = 0.44$–0.58; AEGIS outperforms on larger grids ($\tau = 0.62$–0.67 on case57/118) by capturing nonlinear AC effects. LODF computes in < 0.13 s (analytic formula from line reactances); AEGIS requires training (2–10 s) plus S_c computation (0.2–13 s), totaling 2–23 s per case (case300: 23 s with 200-node subgraph). Brute-force N-1 takes 0.1–2 s. AEGIS is thus slower than LODF but provides richer output (vulnerability spectrum, SVD attacks, sensitivity radii) without requiring line-impedance data.

N-2 contingency connection. The leading singular vector v_1 of S_c naturally identifies multi-edge vulnerabilities. Extracting the top-5 edges from $|v_1|$ and checking their pairwise impact against brute-force N-2 ground truth: edge-level overlap is 40–64% (edges from the SVD direction appear in the ground-truth critical N-2 pairs), with stronger overlap on larger grids (case118: 64%). Pair-level overlap is lower (7–18%) because the SVD direction captures the *most damaging direction* rather than specific pairs, but the constituent edges are individually critical.

Rank stability. Pairwise Kendall τ of AEGIS vulnerability rankings across 10 seeds: $+0.40 \pm 0.29$ (case14), $+0.34 \pm 0.24$ (case30), $+0.78 \pm 0.07$ (case57), $+0.67 \pm 0.12$ (case118). Rankings are more stable on larger grids where the topology is richer. On case118, 60% of the top-10 edges appear in every seed's top-10, and 90% appear in $\geq$80% of seeds.

Binary vs. admittance-weighted adjacency. Binary adjacency outperforms admittance-weighted (P@10 = 0.81 vs. 0.27 on case118). Spearman correlation between admittance magnitude and N-1 severity is near zero across all cases (-0.01 to $+0.15$), confirming that high-admittance lines are not preferentially critical. This reflects the discrete nature of N-1: a line trip fully removes the line regardless of its impedance, so binary sensitivity (uniform per edge) correctly models the all-or-nothing character. Effective-resistance on the binary graph shows moderate correlation with severity ($\tau = +0.49$ on case14/30), consistent with graph-theoretic contingency screening.

Operational caveat. AEGIS is a screening layer, not a standalone contingency tool ($\tau = 0.37$–0.67 is insufficient for direct operational use).

VII. Related Work

We position AEGIS against four research threads.

Adversarial attacks on graph structure. Nettack [1] and Mettack [4] demonstrated structural vulnerability via targeted and global poisoning using GCN [34] surrogates. Subsequent work explored RL-based [2], spectral [43], and bilevel [44] formulations; Wu et al. [45] survey the landscape. These methods use surrogate gradients or heuristic search. AEGIS instead computes the closed-form optimal attack direction via SVD of S_c.

Certified robustness and defense. Convex relaxations [8], [7], randomized smoothing [5], [6], and deterministic certificates [46], [47] provide robustness guarantees but do not identify *which* edges are vulnerable. Empirical defenses [10], [12], [11] harden models without formal vulnerability maps. AEGIS complements both by providing per-edge rankings and per-node radii that reveal the attack surface geometry (section V-B).

Implicit networks and equilibrium analysis. The DEQ framework [16] introduced fixed-point models; IGNN [14] enforces unique equilibria via spectral normalization [18]; EIGNN [48] improves efficiency. For robustness, monotone operator networks [49] provide built-in Lipschitz bounds, and El Ghaoui et al. [24] analyze well-posedness. All address input sensitivity $\|\partial z^*/\partial x\|$, not structural sensitivity $\|\partial Z/\partial A\|$. AEGIS addresses structural sensitivity via the resolvent $(I - J_z)^{-1} J_A$ for implicit models (theorem 1) and the unrolled Jacobian S_K for explicit ones (theorem 5).

Sensitivity analysis and explainability. AEGIS draws on matrix perturbation theory [27], [22]. GNN explanation methods (GNNExplainer [38], PGExplainer [50], gradient-based [37]) optimize for *prediction fidelity*, not adversarial vulnerability: they identify edges important for a correct prediction, while AEGIS identifies edges whose perturbation maximally damages predictions. AEGIS's S_c is perturbation-space optimal by construction (SVD) and provides quantitative sensitivity radii. Our case study (section VI) connects to GNN-based power flow [39], [40], [42] and contingency screening [3].

Over-smoothing and depth. Deep GNNs suffer from over-smoothing [51], which GRAND/GRAND++ [52], [53] address via continuous-depth diffusion. Entezari et al. [54] and Mujkanovic et al. [55] show that many adversarial attacks fail against simple defenses or are artifacts of evaluation. AEGIS sidesteps this debate by analyzing the model's own sensitivity surface rather than searching for attacks via surrogates.

VIII. Conclusion

AEGIS provides a principled pre-deployment diagnostic for GNNs on safety-critical graphs: from a single S_c computation, practitioners obtain the optimal attack direction, per-edge vulnerability rankings, and per-node tolerance budgets. Predictions are tight across 7 architectures and 9 datasets (tightness 1.00 ± 0.01 at $\varepsilon = 0.01$), and the vulnerability spectrum transfers to power grid contingency analysis (P@10 = 0.66–0.81). The per-edge ranking directly informs defense design (section V-G).

Limitations. (1) First-order sensitivity radii are locally tight but not global certificates without second-order bounds. (2) Continuous edge-weight perturbations; discrete insertions/deletions are outside the formal guarantee, though continuous S_c rankings transfer well to discrete removal ($\tau = +0.22$–$+0.54$; table VIII). (3) Standard GAT requires edge-weighted modification for continuous sensitivity. (4) Power flow $\tau = 0.37$–0.67 requires verification for operational use [41]. (5) The matrix-free pipeline's Neumann series requires $\kappa < 1$; for explicit GNNs without contractivity, the dense path ($N \leq 200$) or finite-difference columns are used. (6) IGNN accuracy (77.5% on Cora) lags behind state-of-the-art explicit GNNs ($\sim$82% APPNP); this reflects the accuracy–

guarantee tradeoff inherent in contractive architectures, not a limitation of AEGIS itself. The S_c framework applies identically to higher-accuracy explicit models, albeit without the formal regime characterization (theorem 1).

Ethical considerations. AEGIS identifies optimal attack directions and structurally vulnerable edges, which could be misused to attack deployed GNNs in safety-critical domains. We argue the diagnostic value outweighs this risk: defenders must understand their system's attack surface to protect it, and the same vulnerability map directly informs defense design (section V-G). We will release code with responsible-use guidelines, consistent with established practice in adversarial ML [45].

Future work. Second-order bounds for formal certificates; discrete perturbation models; integration with defense mechanisms [12], [11] for vulnerability-aware training; extending full-graph analysis beyond the current validated boundary (N=7,650 on a single GPU) via distributed JVP computation.

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